Md. Mursalin



Prospective Ph.D. Applicant (Fall 2027) | Multimodal Learning & Trustworthy Medical AI

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GitHub • Thesis Code • Portfolio • LinkedIn • Codeforces

7 research manuscripts • 8 projects built • 1,000+ problems solved • Codeforces Pupil

Research Summary

My work asks one question: **when does a reported score in multimodal or medical AI reflect real signal, and when does it reflect shortcuts, leakage or acquisition artifacts?** I have studied it in depression detection (face, audio, text), retinal foundation models, biometric fusion and plant-disease recognition, using cross-modal attention fusion, ablations, fairness and explainability analysis, and cross-dataset transfer. Alongside research I build complete systems and compete in programming contests, so I can take an idea from experiment to working code. I want to bring this to multi-modal biomedical data and reliable clinical AI.

Education

B.Sc. in Computer Science and Engineering — Netrokona University *Mar 2022 — 2026*
CGPA 3.14 / 4.00 (up to 7th semester; final result to be published) • Thesis supervisor: Md Shovon, Lecturer, Dept. of CSE

HSC — Govt. Laboratory College, Dhanmondi, Dhaka • **GPA 5.00 / 5.00** *2020*

SSC — Ahmed Bawany School and College, Dhaka • **GPA 5.00 / 5.00** *2017*

Research Experience

B.Sc. Thesis: Validity Audit of Tri-Modal Depression Detection on DAIC-WOZ *PyTorch, cross-modal attention*

- Built a cross-modal attention fusion model over facial, audio and text signals, then audited its own pipeline for sentinel artifacts and interviewer shortcuts and tested cross-corpus transfer.
- 9 research phases and 35 training runs with modality ablations, a fairness audit and explainability analysis. [Code]

Confidence-Gated Face–Fingerprint Fusion for Attendance Verification *[Under Peer Review]*

- *SN Computer Science* (SNCS-D-26-06174). Feature-level fusion benchmark under systematic degradation (LFW, SOCOFing, FVC2002/2004), multi-seed: **0.984 AUC** vs. 0.758 face-only and 0.962 fingerprint-only.

Tight Information Hierarchies for Reference-Label Model Selection *[Under Peer Review]*

- Co-author (2nd of 5; with M. Shovon, F. S. Prity, M. A. I. Niloy, M. Z. Rahman). Proves the exact worst-case factor $\lceil r/(k-1) \rceil$ for how much released model-comparison tables reveal about reference-label sensitivity; checked on 1,080 synthetic and ten OpenML audits.

Geographic Generalisation Gap in Retinal Foundation Models *[Under Peer Review]*

- Decomposed the geographic generalisation gap into imaging-device and population effects across five countries.

CLEAR-Papaya: Leakage-Audited, Calibration-Aware Papaya Leaf Disease Recognition *[With Editor]*

- Edge-deployable framework under field-realistic distribution shift. Supervisor: Prof. Jahidur Rahman, Jahangirnagar Univ.

Standardizing Agentic AI: PRISMA Systematic Review (Model Context Protocol) *[Major Revision Submitted]*

- Interoperability, memory and multi-agent coordination: 472 records screened, 230 references verified.

Explainable AI in Healthcare: A Comprehensive Review *[Under Peer Review]*

- *SN Computer Science* (SNCS-D-26-06324). Methods, applications, challenges and future directions.

No manuscript above is published yet.

Selected Projects

- **AEGIS-SIM** (C#, .NET 10, WPF/MVVM, EF Core): desktop logistics simulator with multi-vehicle dispatch, routing and saved-run replay.
- **DocMind AI** (Python, FastAPI, ChromaDB): RAG document Q&A (source). • **Text Summarizer** (Hugging Face, FastAPI): transformer NLP web app.
- **AI Business OS** (n8n, Supabase, Vapi): multi-agent AI receptionist platform (source). • **WhatsApp AI Agent** (Node.js): context-aware agent in a multi-agent suite.
- **Book Recommendation System** (scikit-learn): collaborative and content-based filtering (source).
- **Digital Library System** (MERN): role-based platform, 2nd Place, departmental Innovation Showcasing. • **Office Management System** (Java, JDBC, MySQL).

Competitive Programming, Teaching & Skills

- **Codeforces Pupil (1303)**, **1,000+ problems solved** on Codeforces, Toph and other judges.
- **Rank 51**, DUET IUPC 2026 • **9th Place**, NeU Mymensingh Regional IUPC 2025 (team leader) • **Rank 67**, North Western University IUPC 2023 • **2nd Place**, Innovation Showcasing Competition.

Teaching: Programming Team Coach, Netrokona University (2023–2026) • Technical Educator, YouTube @codewithmursalin • Physics Teacher, UCC (2023–2024).

Research: PyTorch, Hugging Face Transformers, multimodal fusion, explainability, fairness and shortcut audits, multi-seed experiments, PRISMA, LaTeX. **Code:** Python, C/C++, Java, C#, JavaScript, SQL, Git, Linux.

Languages: Bengali (native), English (professional working proficiency); IELTS not yet taken, preparing before the application deadline.

References

Md Shovon — Lecturer, Dept. of CSE, Netrokona University (B.Sc. thesis supervisor) • mdshovon@neu.ac.bd • 01768668915
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